Recurrent Memory Lab

How a fixed-size state can learn from demonstrations - and how it can forget

1. The idea in one sentence

A recurrent state can absorb a growing stream of demonstrations without growing its shape, but compression makes interference and forgetting possible.

2. Start with a learner-visible substrate

The demo is deliberately tiny. A 5 x 5 matrix is the memory. Each demonstration writes a key/value association by an outer product; each query projects a key through the accumulated matrix. The point is not to make a powerful model. The point is to expose the variable that normally disappears inside a large neural network: *state*.

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Teaching equation / evidence boundary: M_t = lambda M_(t-1) + alpha k_t v_t^T (toy
only). Official BDH uses richer neuron/synapse dynamics; BDH-CQ is the published
recurrent-latent model described in source [2].
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3. The live experiment

The page opens with a clean preset, so the learner immediately sees a correct prediction. Adding demonstrations changes the state but not its dimensions. A similarity slider makes distinct keys share representational features. A conflict button adds a wrong association for the query key. The output shows ground truth, prediction, and the score vector side by side. This turns the claim into something the learner can challenge.

4. Why fixed size matters

A token-by-token cache stores a growing record of past representations. A recurrent memory instead carries forward a summary. This can reduce the amount of inference-time memory that must grow with sequence length, but the summary is necessarily lossy in some tasks. The educational interface makes the trade-off concrete: five columns remain five columns while more evidence is written.

5. Why interference matters

Suppose two keys are orthogonal. A write to one key does not affect the other in the idealized toy. Now make their vectors similar. The outer-product writes share representational directions, so reading one key can recover value components contributed by another. This is the same kind of capacity/crosstalk question that appears whenever many experiences are compressed into one evolving state.

6. BDH: from matrix toy to synaptic dynamics

Dragon Hatchling (BDH) is a substantially richer research architecture. Its paper describes a scale-free graph of locally interacting neuron particles and reports that working memory during inference relies on synaptic plasticity with Hebbian learning. The paper also studies sparse positive activations and interpretable state. The correct pedagogical bridge is therefore: the toy makes “memory as evolving internal state” visible; BDH provides a concrete frontier architecture in which memory is implemented through learned local dynamics. The toy is not BDH and should never be presented as a faithful reimplementation.


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7. BDH-CQ: the in-context-learning bridge

BDH-CQ is especially aligned with the chosen topic. Its 2026 technical report describes inference-time inputs continuously updating recurrent memory, after which a query is solved through iterative computation in a high-dimensional latent space without verbalizing intermediate reasoning. This gives the learner a clean conceptual picture: demonstrations update internal state; state supports the query; additional latent computation can refine the solution. The artifact teaches the first two pieces directly and uses the paper to explain the third, without pretending the browser toy reproduces the 150M-parameter model.


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8. Frontier context

The concept is active well beyond BDH. Griffin combines gated linear recurrences with local attention. RecurrentGemma packages the Griffin architecture into open language models and emphasizes the efficiency of a fixed-sized recurrent state. Titans introduces a learned neural long-term memory for historical context and reports experiments on very long contexts. These systems make the same design pressure visible from different directions: keep useful information available without paying the full cost of carrying every prior token through every future operation.

9. What the demo does not prove

The demo does not prove that recurrent memory is better than Transformer attention, that fixed-size state is sufficient for arbitrary tasks, or that BDH-CQ performance follows from the toy equation. It does prove something narrower: under the toy dynamics, the state shape stays fixed while demonstrations accumulate, and controlled similarity/conflict can change retrieval accuracy. That narrow claim is exactly why the experiment is useful.

10. 60-second lesson

Load the preset. Query B. Confirm the truth. Increase key similarity. Add a conflicting demonstration. Query B again. Then explain: (a) what stayed fixed, (b) what changed, (c) why the estimate shifted, and (d) how that resembles but does not equal recurrent memory in BDH-CQ. A successful lesson is not “recurrent models are good”; it is a mechanistic explanation of a measurable trade-off.

Research notes and source map

Claim used in this blog	Primary source
BDH uses synaptic plasticity / Hebbian working memory and a brain-inspired graph	[1] BDH paper
BDH-CQ updates recurrent memory at inference and performs latent iterative reasoning	[2] BDH-CQ paper
Gated linear recurrences and local attention can support efficient language models	[3] Griffin
RecurrentGemma is an open model family built on Griffin and uses fixed-sized state	[4] RecurrentGemma
Neural long-term memory can learn to memorize historical context at test time	[5] Titans
Pathway frames local neuron/synapse dynamics as the basis for memory and reasoning	[6] Equations of Reasoning

Full references

[1] Kosowski, A. et al. (2025). The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. arXiv:2509.26507. https://arxiv.org/abs/2509.26507

[2] Engdahl, B. et al. (2026). BDH-CQ: In-Context Learning with Recurrent Latent Reasoning. arXiv:2608.09888. https://arxiv.org/abs/2608.09888

[3] De, S. et al. (2024). Griffin: Mixing Gated Linear Recurrences with Local Attention for Efficient Language Models. arXiv:2402.19427. https://arxiv.org/abs/2402.19427

[4] Botev, A. et al. (2024). RecurrentGemma: Moving Past Transformers for Efficient Open Language Models. arXiv:2404.07839. https://arxiv.org/abs/2404.07839

[5] Behrouz, A., Zhong, P., Mirrokni, V. (2025). Titans: Learning to Memorize at Test Time. arXiv:2501.00663. https://arxiv.org/abs/2501.00663

[6] Pathway (2026). The Equations of Reasoning. https://pathway.com/research/the-equations-of-reasoning

Implementation and provenance

The browser artifact is original JavaScript/HTML/CSS with no third-party libraries. The synthetic key/value examples are original. AI assistance was used for research organization, drafting, and initial code generation; the repository explicitly discloses this and requires human review of every claim and implementation detail. The code is MIT licensed.